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MILITARY SPECIFICATION

COPPER PRESSURE CYLINDERS AND COPPER PRESSURE SPHERES, CANNON (MAJOR AND MEDIUM), CANNON (MINOR), SMALL ARMS, AND PISTOL AND REVOLVER

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Army 52-1-51A
15 August 1944

This specification consists of this cover sheet and Specification 52-1-51A, dated 15 August 1944, together with Amendment-2, dated 2 December 1947, attached hereto, without modification.

Copies of this specification may be obtained upon application to the Office, Chief of Ordnance, Washington 25, D. C.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

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U. S. ARMY
SPECIFICATION

No. 52-1-51A

15 AUGUST 1944

COPPER PRESSURE CYLINDERS AND COPPER PRESSURE SPHERES, CANNON (MAJOR AND MEDIUM), CANNON (MINOR), SMALL ARMS, AND PISTOL AND REVOLVER

A. APPLICABLE SPECIFICATIONS.

A–1 The following specifications and drawings in effect on the date of invitation for bids are hereby made a part of this specification:

U. S. Army Specifications

No. 100-2 —Standard Specification for Marking Shipments;
No. 57–154–1—Special Copper Rod for Pressure Cylinders, all types.

Ordnance Department Drawings

No. 65–1–18 —Cannon (Major and Medium) Pressure Cylinder;
No. A39436 —Cannon (Minor) and Small Arms Pressure Cylinder;
No. A47753 —Pistol and Revolver Pressure Cylinder;
No. B129626—Label;
No. C44747 —Method of Packing Cylinders, Copper, Pressure.
No. A7675125—Sphere, Pressure.

A–2. The specifications referred to in the specifications listed above, but not listed herein, shall not apply.

B. GRADE, CLASS, AND SIZE.

B–1. This specification covers one grade of copper pressure cylinders and copper pressure spheres, primarily intended for use in mensuring pressures of ammunition, in the following sizes and classes:

B–1a. Cylinders, class 1 (annealed):

Size A, for cannon (major and medium).
Size B, for cannon (minor).
Size C, for small arms.
Size D, for pistol and revolver.

B–1b. Spheres, class 1 (annealed) or class 2 (hard), as specified:

Size A, for small arms.
Size B, for special tests.

Hard spheres will be annealed by Frankford Arsenal or other Government agency purchasing the spheres.

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C. MATERIAL AND WORKMANSHIP.

C-1. The material used for the manufacture of the copper pressure cylinders and copper pressure spheres shall meet the requirements of U. S. Army Specification 57–154–1.

C-2. All copper pressure cylinders and copper pressure spheres shall have a clean, smooth surface and be free from flaws, seams, pipes, dents, cuts, burrs, and any or all other mechanical defects which may affect the accuracy of pressure measurements obtained therefrom.

D. GENERAL REQUIREMENTS.

D-1. There are no general requirements applicable to this specification.

E. DETAIL REQUIREMENTS.

E-1. The copper pressure cylinders, after annealing, shall have the finished dimensions as follows:

Size A.—Cannon (major and medium):
Average diameter 0.2580–0.0020 in.
Average length 0.5010–0.0020 in.
Size B.—Cannon (minor):
Average diameter 0.2285–0.0020 in.
Average length 0.4010–0.0020 in.
Size C.—Small arms:
Average diameter 0.2285–0.0020 in.
Average length 0.4010–0.0020 in.
Size D.—Pistol and revolver:
Average diameter 0.1465–0.0020 in.
Average length 0.4010–0.0020 in.

The standard deviation of the lengths of any single lot of cylinders shall not exceed 0.00020 inch; the standard deviation of the diameters shall not exceed 0.00050 inch; and the faces of the individual copper pressure cylinders shall be parallel to each other within 0.0005 inch.

E-2. The copper pressure spheres shall have finished diameters as follows:

A.—Small arms 0.3115–0.3135 inch;
B.—As may be specified in the contract.

The standard deviation of the diameters shall not exceed 0.00020 inch.

E-3. *Annealing.*—The copper pressure cylinders or copper pressure spheres, class 1, shall be annealed uniformly; and after annealing, the physical state of the copper cylinders shall be such that, upon application of an initial load of 45,000 pounds per square inch of the face of the cylinder, the resulting remaining length shall be between 0.81 and 0.84 of the original length; the physical state of the copper spheres, class 1, shall be such that upon application of a load of 28,500 pounds per square inch of the cross section of the sphere, the distance between the resulting flattened faces shall be between 0.65 and 0.75 of the original diameter.

E-4. *Uniformity.*—The standard deviation of all the sets obtained in the construction of the tarage table shall not exceed 0.0010 inch.

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E-5. The contractor shall furnish the Government inspector the record, in triplicate, of the manufacturer and the date of manufacture of the copper rod from which the copper pressure cylinders or spheres are made, together with the date of manufacture of the copper pressure cylinders or spheres, and the date, duration and temperature of their anneal.

E-6. *Tarage table.*—The appropriate tarage table or tables for reading the pressures shall be made by Frankford Arsenal.

F. METHODS OF INSPECTION AND TEST.

F-1. Unless otherwise specified, inspection and tests shall be made at the place of manufacture.

F-2. *Size of lots.*—Unless otherwise specified, the number of copper pressure cylinders or spheres comprising a single lot offered for acceptance, shall be at least 50,000, less the number rejected for visual or gaging defects, or both, by the Government inspector. All copper pressure cylinders or spheres in one lot shall be made from copper rod from one billet and shall be of the same size.

F-3. *Visual inspection and gaging.*—

F-3a. *Visual inspection.*—Each copper pressure cylinder in the lot shall be inspected visually, and those not conforming to the requirements of C-2, shall be discarded.

F-3b. *Gaging.*—The sample for gaging and for construction of the tarage table shall be selected as follows:

	Size of sample (cylinders)
Copper pressure cylinders:	
Size A.—Cannon (major and medium)	300
Size B.—Cannon (minor)	450
Size C.—Small arms	450
Size D.—Pistol and revolver	350
Copper pressure spheres, class 1 or class 2:	*Spheres*
Size A.—Small arms	450
Size B.—As may be specified in the contract.	

F-3b (1). Each cylinder in the sample chosen shall be identified and then measured for length, diameter, and parallelism of the bases. The average and the standard deviation of the length and of the diameter shall then be computed. If the results comply with the requirements of paragraph E-1, the sample shall be used for the construction of the tarage table; if not, the lot shall be returned to the contractor for correction.

F-3b (2). Each sphere in the sample chosen shall be identified, and then measured for diameter. The average and the standard deviation of the diameter shall then be computed. If the results comply with the requirements of E-2, in the case of class 1 spheres, the sample shall be used for the construction of the tarage table; in the case of class 2 spheres the lot shall then be accepted. If the results do not comply with the requirements of E-2, the lot shall be returned to the contractor for correction.

F-3c. If the sample selected under paragraph F-3b above comply with the requirements of E-1 or E-2, as the case may be, it shall be

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sent to Frankford Arsenal, Small Arms Ammunition Division, Engineering Branch, for construction of the tarage table, except in the case of class 2 spheres.

F–3d. In the case of class 2 spheres, if the sample selected under F–3b above comply with E–2, the entire lot shall be shipped as specified in the contract.

F–4. *Tarage table and compression curve.*—

F–4a. The samples delivered under F–3c above shall be divided into groups of 20 coppers each, as follows:

	Number of groups of 20 coppers
Copper pressure cylinders:	
Size A	12
Size B	20
Size C	20
Size D	15
Copper pressure spheres:	
Size A	20
Size B	As may be specified in the contract.

The remaining coppers in the sample may be used as spheres, necessary for adjusting the machine. Each copper shall be measured before and after being compressed. If the individual coppers remain identified, the measurements obtained under paragraphs F–3b (1) and F–3b (2) shall be used; otherwise, a new measurement must be made.

F–4b. The coppers shall then be compressed in the Frankford Arsenal compression machine, 1 group of 20 coppers at each of the loads given in table I. The average set and the standard deviation shall be determined for each load. If the results thus obtained comply with the requirements of E–3 and E–4, the lot shall be accepted; if not the lot shall be returned to the contractor for reannealing.

NOTE.—In general, the points listed in table I hereafter are sufficient for constructing a compression curve. If, however, the curve thus obtained include too many points of inflection, or if it be otherwise unsatisfactory, as many more such points as may be desired may be determined and used in constructing the compression curve and the tarage table.

F–4b (1). These points having been thus determined, the intermediate points shall be obtained by interpolation at each 200 pound per square inch over the entire range of observed points. These figures, when tabulated, shall constitute the tarage table.

F–4b (2). Each tarage table shall show:

Rod:
- The manufacturer.
- The year of manufacture.

Copper pressure cylinders, and copper pressure spheres:
- The manufacturer.
- The date or dates of annealing.
- The average length, uncompressed.
- The date or dates on which the compressions were made.
- The number of the drawing showing the compression curve.

F–4b (3). The compression curve shall show the same data as the tarage table.

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F–5. *Retests.*—If a lot minus the coppers discarded under F–3a, be returned to the contractor, as provided in F–3b (1), F–3b (2), or F–4b, and if the lot, minus those discarded under F–3a be resubmitted by the contractor, the visual inspection in the retest shall be limited to the detection and removal of coppers having burrs, etc., caused by correcting the lengths of the copper pressure cylinders; the samples for gaging and for the tarage table shall be the same size as for the first test.

TABLE I

Cannon (major and medium)		Cannon (minor) and small arms		Pistol and revolver	
Piston area ⅒ sq. in.		Piston area ⅕ sq. in.		Piston area ⅛ sq. in.	
Applied load	Corresponding chamber press	Applied load	Corresponding chamber press	Applied load	Corresponding chamber press
lb.	*lb./sq. in.*	*lb.*	*lb./sq. in.*	*lb.*	*lb./sq. in.*
500	5,000	300	9,000	100	8,000
1,000	10,000	500	15,000	150	12,000
1,500	15,000	700	21,000	200	16,000
2,000	20,000	900	27,000	250	20,000
2,500	25,000	1,000	30,000	300	24,000
3,000	30,000	1,100	33,000	350	28,000
3,500	35,000	1,200	36,000	400	32,000
4,000	40,000	1,300	39,000	450	36,000
4,500	45,000	1,400	42,000	500	40,000
5,000	50,000	1,500	45,000	600	48,000
5,500	55,000	1,600	48,000	700	56,000
6,000	60,000	1,700	51,000	750	60,000
		1,750	52,500	800	64,000
		1,800	54,000	900	72,000
		1,850	55,500	1,000	80,000
		1,900	57,000		
		2,000	60,000		
		2,200	66,000		
		2,400	72,000		
		2,500	75,000		

G. PACKING AND MARKING.

G–1. *Packing.*—The copper pressure cylinders, with appropriate tarage tables which shall be furnished by the contracting officer, shall be packed in accordance with Ordnance drawing C14747, and each box shall be labeled in accordance with Ordnance drawing B120823.

G–2. *Marking.*—The packing boxes containing the copper pressure cylinders shall be marked in accordance with Ordnance drawing B120826.

H. NOTES.

H–1. The use of this specification, whenever applicable, is mandatory on the procuring agencies of the Army.

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NOTE.—Copies of this specification may be obtained from Army Service Forces, Office, Army Service Forces, Washington, D. C.

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